

Detailed Report

Preliminary Conceptual Design of a Ground-Based LEO Tracking Radar

Executive Summary

This report documents the engineering approach used to develop a preliminary conceptual design for a ground-based monostatic radar capable of detecting and tracking small LEO objects between 500 km and 800 km altitude. The design target is a metallic spherical object of approximately 10 cm diameter. The assessment requirements include detection, acquisition, continuous tracking, range estimation, radial velocity estimation, angular position estimation and orbit-determination support.

The problem was approached as a system-engineering exercise rather than as an attempt to produce a production-ready radar. The work was decomposed into requirements, frequency selection, waveform design, link budget, antenna, transmitter, receiver, signal processing, tracking, orbit determination and sensitivity analysis. The resulting preliminary architecture uses 3 GHz operation, a 15 MHz LFM waveform, a 16 m parabolic reflector and 15 kW peak RF power.

At the 800 km design point, the preliminary link budget gives approximately 15.2 dB ideal processed SNR. After allocating 5 dB for implementation and processing losses, approximately 10.2 dB effective margin remains. The 15 MHz waveform provides 10 m theoretical range resolution and approximately 0.094 m/s theoretical velocity resolution for the assumed coherent processing interval.

1. Problem Definition and Requirements

The assignment asks for a ground-based radar capable of detecting and tracking LEO objects from 500 km to 800 km altitude, including a target with an approximate 10 cm diameter metallic-sphere RCS. The system must support detection, acquisition, continuous tracking, range, radial velocity, angular position and orbit-determination support.

The principal numerical requirements are: detection margin ≥ 10 dB, range resolution ≤ 10 m, and velocity accuracy < 1 m/s. The 800 km case is treated as the primary design point because it is the most demanding range.

2. Engineering Approach

The design process followed a top-down decomposition:

Requirements -> RF frequency -> waveform -> radar equation/link budget -> antenna -> transmitter -> receiver -> digital processing -> detection -> Doppler -> tracking -> orbit determination -> sensitivity analysis.

This decomposition helped prevent individual component choices from being made independently of system performance. Each major parameter was linked back to a requirement or to a system-level trade-off.

3. Frequency Selection

A preliminary operating frequency of 3 GHz was selected. At this frequency, $\lambda = 0.1$ m. The choice represents a compromise between antenna aperture, angular resolution, Doppler sensitivity, RF technology practicality and atmospheric propagation. A detailed site-specific propagation analysis would be required in a subsequent design phase.

The 10 cm wavelength also gives a direct Doppler relationship of $f_D = 2v_r/\lambda = 20v_r$, which provides useful sensitivity to radial velocity.

4. Waveform Design

An LFM chirp was selected because it allows a relatively long, high-energy pulse to be transmitted while achieving fine range resolution after matched filtering. The baseline waveform uses $B = 15$ MHz and $\tau = 100$ μ s.

The time-bandwidth product is $BT = 1500$. The theoretical range resolution is $\Delta R = c/(2B) = 10$ m. The ideal pulse-compression gain is $10\log_{10}(1500) \approx 31.8$ dB.

This choice demonstrates the interaction between waveform duration, bandwidth, transmitted energy and range resolution rather than treating range resolution as an isolated specification.

5. Initial Link Budget

For a 10 cm diameter metallic sphere, a nominal RCS of $\sigma = \pi(0.05)^2 = 0.00785$ m² is used. This is an idealized assessment assumption; the RCS of a real object may vary significantly with geometry, attitude and frequency.

The monostatic radar equation is used to estimate the received echo. With $P_t = 15$ kW, $G_t = G_r \approx 52.2$ dBi, $\lambda = 0.1$ m, $\sigma = 0.00785$ m² and $R = 800$ km, the estimated received power is approximately -135.8 dBm.

For a 15 MHz bandwidth and 3 dB receiver noise figure, the receiver noise estimate is approximately -99.2 dBm. The resulting raw SNR is approximately -36.6 dB. This shows that detection cannot depend on instantaneous SNR and motivates pulse compression and coherent processing.

6. Antenna Design

A 16 m parabolic reflector is proposed. With an assumed 65% aperture efficiency, $G = \eta(\pi D/\lambda)^2$ gives approximately 52.2 dBi. The approximate HPBW is $70\lambda/D \approx 0.44^\circ$.

A two-axis azimuth/elevation mount is proposed. For angular tracking, a monopulse architecture is preferred because it produces instantaneous angular-error information relative to boresight. A preliminary $\leq 0.05^\circ$ measurement target is retained as a design target requiring detailed antenna/feed and calibration verification.

7. Transmitter Design

The preliminary peak RF output is 15 kW. A modular solid-state architecture is preferred, using multiple PA modules with power combining. A 60% PA efficiency is used only as a preliminary assumption for system power and thermal estimates.

The custom engineering challenges are expected to include power combining, thermal management, impedance matching, monitoring, protection, redundancy and integration with the T/R network. Individual RF amplifier, driver and protection components can potentially be selected from COTS technology.

8. Receiver Design

The proposed receiver chain is: antenna -> T/R protection -> filtering -> LNA -> mixer/downconversion -> IF filtering/amplification -> ADC -> FPGA/DSP. A 3 dB receiver noise figure is assumed for the conceptual link budget.

The receiver must recover from high-power transmission, preserve coherent phase information and provide sufficient dynamic range for weak echoes. Detailed T/R protection, blanking and recovery-time design remain future work.

9. Signal Processing

The digital processing chain is: digital down conversion -> matched filtering/pulse compression -> range processing -> coherent integration -> Doppler FFT -> range-Doppler map -> CFAR -> detection. This processing transforms the weak raw echo into a detectable range-Doppler response.

With 100 coherent pulses, ideal coherent integration gain is 20 dB. Combined with approximately 31.8 dB pulse-compression gain, the ideal processed SNR is approximately 15.2 dB. Allocating 5 dB for implementation and processing losses leaves approximately 10.2 dB effective preliminary detection margin.

10. Doppler, PRF and Velocity

For $R_{\max} = 800$ km, the simple unambiguous-range constraint gives $\text{PRF} \approx 187.5$ Hz. With 100 pulses, the reference CPI is approximately 0.533 s. The theoretical velocity resolution is therefore approximately 0.094 m/s.

However, the low PRF produces Doppler ambiguity for LEO velocities. This is a significant system-level issue identified during the design process. The proposed solution is staggered/multiple-PRF operation and ambiguity-resolution processing. The exact PRF set should be optimized in detailed waveform scheduling.

This illustrates an important design trade-off: maximizing unambiguous range and maximizing unambiguous Doppler cannot be achieved simultaneously with a single simple PRF.

11. Detection and Tracking

Detection is followed by track initiation, range/Doppler/angle measurement, filtering and prediction. A closed-loop Az/EI servo uses predicted target position and angle error to keep the antenna pointed at the object.

The tracking architecture is therefore: detection -> acquisition -> measurement -> tracking filter -> prediction -> servo control -> antenna -> next measurement.

12. Orbit Determination Support

The radar provides time-tagged measurements $z = [R, \text{range-rate}, \text{Az}, \text{EI}]$. Repeated observations can be transformed into an Earth-referenced coordinate system and used to estimate the six-state vector $[x, y, z, v_x, v_y, v_z]$. An Extended Kalman Filter is proposed as a preliminary state-estimation architecture.

The design does not assume that one observation determines a complete orbit. Instead, a time history of measurements over an observation arc supports prediction and correction of the estimated state.

13. Sensitivity Analysis and Design Trades

<i>Variation</i>	<i>Approx. margin</i>
<i>Baseline</i>	10.2 dB
<i>Range = 700 km</i>	12.5 dB
<i>Range = 500 km</i>	18.4 dB
<i>RCS = 0.5× nominal</i>	7.2 dB
<i>RCS = 2× nominal</i>	13.2 dB
<i>Tx power = 10 kW</i>	8.4 dB
<i>Tx power = 20 kW</i>	11.5 dB
<i>Antenna = 14 m</i>	8.1 dB
<i>Antenna = 18 m</i>	12.2 dB

The sensitivity study shows that range is especially important because received power varies approximately as R^{-4} . For a monostatic system using the same aperture for transmit and receive, received power scales approximately as D^4 . RCS uncertainty is also significant. Increasing transmitter power improves margin, but the improvement is logarithmic and comes with additional thermal and hardware complexity.

The baseline 10.2 dB effective margin is therefore better interpreted as a conceptual threshold result than as a large operational reserve.

14. Custom vs COTS Engineering

The conceptual architecture intentionally separates items that can reasonably be based on COTS technology from subsystems that require system-specific engineering.

<i>Subsystem</i>	<i>Approach</i>
<i>RF synthesizer / exciter</i>	COTS + custom control
<i>ADC / FPGA</i>	COTS hardware + custom firmware
<i>LNA / mixer / IF</i>	COTS candidates
<i>PA modules</i>	COTS candidates
<i>15 kW PA assembly</i>	Custom integration
<i>Power combiner / T-R network</i>	Custom high-power integration
<i>16 m reflector / feed</i>	Custom
<i>Monopulse feed</i>	Custom
<i>Az/EI mount</i>	Custom integration
<i>Pulse compression / Doppler / CFAR</i>	Custom DSP/software
<i>Tracking / orbit estimation</i>	Custom software
<i>Thermal and timing distribution</i>	Custom system integration

15. Limitations and Further Work

- Detailed electromagnetic modelling and antenna/feed pattern characterization.
- Site-specific atmospheric propagation and weather-loss analysis.
- Detailed clutter, interference and false-alarm analysis.
- Optimization of staggered/multiple PRFs for range-Doppler ambiguity resolution.
- Tracking-loop dynamics, maximum angular rate and servo acceleration analysis.
- Detailed high-power PA, combining and thermal design.
- Hardware-level T/R protection and receiver recovery analysis.
- Detailed orbit-determination performance using representative trajectories and measurement errors.
- Hardware component selection, availability, reliability and cost assessment.

16. Engineering Reflection

This assessment was approached as a first complete end-to-end radar system design exercise. Rather than beginning with a fixed hardware solution, the problem was progressively decomposed into smaller engineering questions and each assumption was checked against its effect on the overall system.

The design process highlighted that radar performance is strongly coupled: frequency affects wavelength and Doppler; bandwidth affects range resolution; antenna aperture affects gain and beamwidth; PRF creates a range-Doppler trade-off; and target RCS and range strongly influence detection margin. Working through these interactions was an important part of arriving at the final architecture.

The resulting design is intentionally presented as a preliminary conceptual architecture rather than a production-ready system. Several areas remain open for detailed engineering and validation. The main outcome of the exercise is therefore not only the proposed set of parameters, but also the demonstrated process of decomposing an unfamiliar problem, identifying governing relationships, evaluating trade-offs and identifying where further engineering is required.

17. Final Conclusion

The proposed 3 GHz, 15 kW, 16 m monostatic radar provides a traceable preliminary architecture for detection and tracking of a 10 cm-class LEO object at ranges up to 800 km. Under the stated assumptions, the design provides 10 m theoretical range resolution, approximately 0.094 m/s theoretical velocity resolution and approximately 10.2 dB effective detection margin at the 800 km design point.

The sensitivity analysis identifies target RCS, range and antenna aperture as important performance drivers. Further optimization is required to increase operational margin and to validate antenna, thermal, tracking, propagation and orbit-determination performance.

The design should therefore be regarded as a system-level starting point from which a detailed radar implementation could be developed.